

stacked in his favor or against him. Even when the odds are clearly stacked against the gambler, he sometimes transcends the odds by believing that destiny selected him in some manner. This shows in the very sophisticated people one meets in casinos where they normally should not be found. I even ran into world-class probability experts who had a gambling habit on the side, throwing all of their knowledge to the wind. For example, a former colleague of mine and one of the most intelligent people I have ever met, frequently went to Las Vegas, and seemed to be such a turkey that the casino provided him with complimentary luxury suites and transportation. He even consulted a fortune teller prior to taking large trading positions and tried to get reimbursed by our employer.

The Skinner Pigeon Experiment

At 25, I was totally ignorant of the behavioral sciences. I had been fooled by my education and culture into believing that *my superstitions were cultural*, and that, consequently, they could be shed through the exercise of so-called reason. Taken at the general level of society, modern life would eliminate them as science and logic would enter it. But in my case, I was over time getting more sophisticated intellectually, but the floodgates of randomness were bursting and I was becoming more superstitious.

These superstitions needed to be biological – but I was brought up in an era when the dogma was that it was nurture, rarely nature, that was the culprit. Clearly, there was nothing cultural about my link between my wearing glasses and a random market outcome. There was nothing cultural in my link between my use of entrance and my performance as a trader. There was nothing cultural in my wearing the same tie as the day before. Something in us has not developed properly over the past thousand years and I was dealing with the remnant of our old brain.

To probe the point further, we need to look at such formations of causal associations in the lower forms of life. The famous Harvard psychologist B.F. Skinner constructed a box for rats and pigeons,

equipped with a switch that the pigeon can operate by pecking. In addition, an electrical mechanism delivers food into the box. Skinner designed the box in order to study more general properties of the behavior of a collection of nonhumans, but it was in 1948 that he had the brilliant idea of ignoring the lever and focusing on the food delivery. He programmed it to deliver food at random to the famished birds.

He saw quite astonishing behavior on the part of the birds; they developed an extremely sophisticated rain-dance type of behavior in response to their ingrained statistical machinery. One bird swung its head rhythmically against a specific corner of the box, others spun their heads anti-clockwise; literally all of the birds developed a specific ritual that progressively became hard-wired into their mind as linked to their feeding.

This problem has a more worrying extension; we are not made to view things as independent from each other. When viewing two events A and B, it is hard not to assume that A causes B, B causes A, or both cause each other. Our bias is immediately to establish a causal link. While to a budding trader this causes hardly any worse costs than a few pennies in cab fare, it can draw the scientist into spurious inference. For it is harder to act as if one were ignorant than as if one were smart; scientists know that it is emotionally harder to reject a hypothesis than to accept it (what is called type I and type II errors) – quite a difficult matter when we have such sayings as *felix qui potuit cognoscere causas* (happy is he who understands what is behind things). It is very hard for us to just shut up. We are not cut out for it. Popper or not, we take things too seriously.

Philostratus Redux

I offered no solution to the problem of statistical inference at a low resolution. I discussed in Chapter 3 the technical difference between noise and meaning – but it is time to discuss the execution. The Greek philosopher Pyrrho, who advocated a life of equanimity and indifference, was criticized for failing to keep his composure during a